
PalmSource™ 99
Creating a
Conduit for Macintosh®
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Overview

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- Synchronization Overview
- Types of Synchronization
- HotSync® Architecture
- Features of the Conduit Development Kit
- Palm OS® Conduit Project Stationery
- Developing a Conduit for Macintosh
- Installing Conduits on Mac OS
- Summary

Synchronization Overview

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- What is Synchronization
  - A two-way synchronization of records between your Palm Computing® platform device and your computer
- Synchronization is performed by HotSync® and your conduit
- Three Types of Synchronization
  - Desktop
  - Network
  - Server

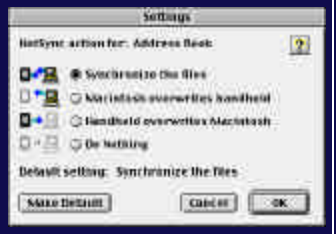
Types of Synchronization

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- Desktop Synchronization
  - Handheld and synchronization engine on local workstation
  - Data usually located on local workstation
- Network Synchronization
  - Handheld connected to any computer on network
  - Synchronization runs on handheld user's workstation
  - Data accessible from handheld user's workstation
- Server Synchronization
  - Handheld connected directly to network (or client)
  - Synchronization runs on servers
  - Data and synchronization engines distributed

Synchronization Dataflow

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Mirror image	Ideal for applications that run on both platforms and allow modifications on both; makes data identical on both platforms. (address book and date book applications)	Your conduit must provide conflict resolution solutions when the same record has been modified on both platforms. This must occur without user intervention.
One directional Upload / Download	For applications that run on both platforms but only allow data modification on one; copies the data to the other platform.	Requires less time, and should be used when possible.
Transaction based	For applications that need to perform additional processing on the desktop computer between record synchronizations. (e-mail, web, PQA, etc.)	Only use this when required, because it slows down the HotSync® process.
Back up	For applications that don't have desktop components; simply copies the handheld data to the desktop.	Can use the Backup conduit provided by Palm Computing.

HotSync® Architecture

HotSync Manager

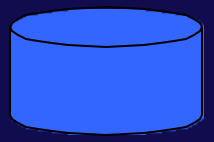


Conduit API
- OpenConduit()
- ConfigureConduit()

Application-Specific Conduit



Data Source



- Local/networked data file
- Local/networked database
- Web-based data source
- E-mail or news server
- Dial-in connection

Conduit Base Classes
- CSynchronizer
- CRecordIterator
- CSyncRecord

SyncManager API
- SyncOpenDb()
- SyncReadNextModified()
- SyncWriteRec()

Handheld



Contains applications/databases
- Unique creator id for each application
- Built-in HotSync device client
- HotSync process initiated by the device

Features of Palm OS® Conduit Development Kit for Macintosh

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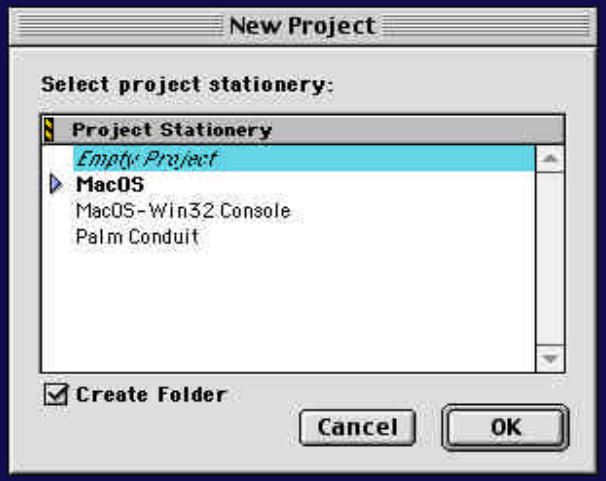
- Palm Conduit Project Stationery
  - Project Starter Code Generation
- Conduit Base Classes
  - Core synchronization logic
- Conduit Sample Projects
  - Source code for Palm standard applications

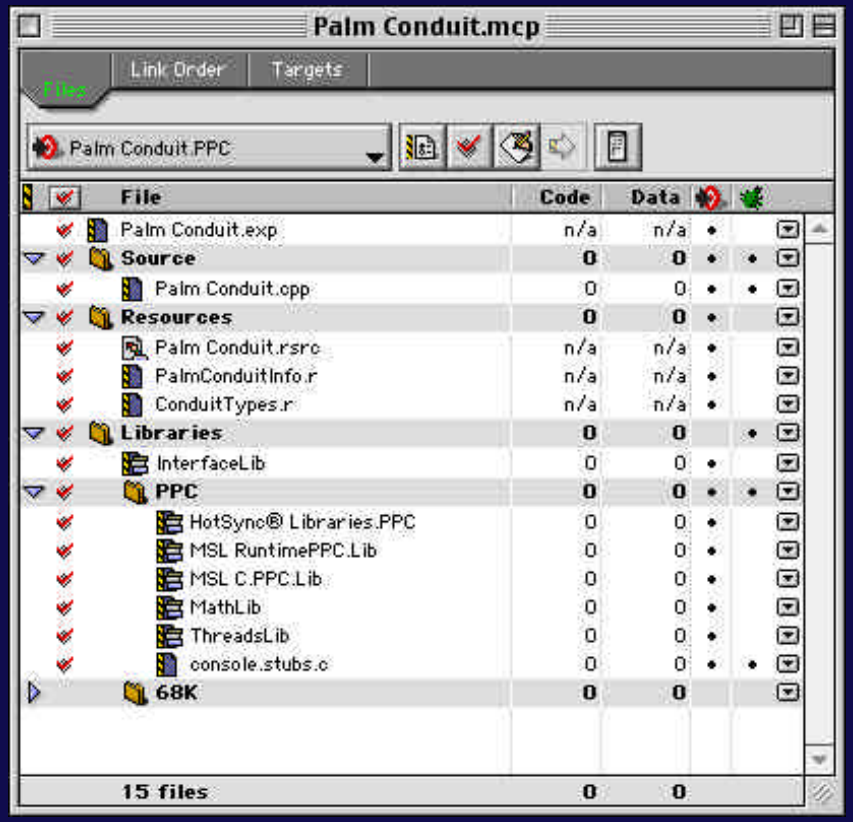
Features of Palm OS® Conduit Development Kits

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- Install Aide API
  - Permits easy installation/management of Palm Computing Platform applications
- SyncManager API
  - Database/record access

Palm OS® Conduit Project Stationery





Easy configuration of conduit starter code

- Metrowerks CodeWarrior ® 3 or 4
- Generates starter code for Mac® OS conduits
- Generic Conduit to be supported in future releases

Conduit Entry Points

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Required

- OpenConduit()
  - Synchronization entry point
- GetConduitName()
  - Returns name of conduit for display purposes
- GetConduitVersion()
  - Returns conduit version
- GetActionString()
  - Returns the string seen by the user in 'Next HotSync® action' column of HotSync® Manager actions dialog
- ConfigureConduit()
  - Presents a dialog that allows the user to configure the conduit

Developing a Simple Conduit

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- Project Stationery
- Modifying the base code
- Options for creating conduits
  - Palm Conduit code
  - Example code

Conduit Installation

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- Install Aid Library
- Useful API functions
  - PltInstallFile / PltInstallFileFS
  - PltRemoveInstallFile / PltRemoveInstallFileFS
  - PltGetUserDirectory / PltGetUserDirectoryFS
- Iteration of all Users
  - PltGetUserCount
  - PltGetUser

Summary

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- The Palm Computing® platform is the leader in handheld computing and information management
- Synchronization (Data Exchange) plays an important role in handheld computing. (personal and enterprise information management)
- Successful implementations depend on choosing the right synchronization strategy
- The HotSync® architecture is adaptable to many synchronization scenarios
- The Palm OS® Conduit Project Stationery provides a flexible framework for development of C++ Conduits for Macintosh
- The Palm Computing® CDK provides a complete set of tools for conduit development for Macintosh


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